

Big Multiplication

Every graded set, every week — 4 weeks, 20 worksheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Multiples of Ten

Get these automatic and the algorithm gets easy.

✦ Warm-Up 6 ITEMS

20×30

40×50

60×70

30×300

80×20

90×40

◆ Core GRADED • SHOW YOUR WORKING

1. 400×60

2. 700×800

3. 25×40

4. 120×50

5. 36×200

★ Challenge NOT GRADED

C1. 250×400

.....

C2. 1250×80

.....

Partial Products

Four boxes, four numbers, one total.

Warm-Up 6 ITEMS

23 × 10

23 × 4

Add 230 and 92

So 23 × 14 is

31 × 20

31 × 2

Core GRADED • SHOW YOUR WORKING

1. 31 × 22

2. 45 × 23

3. 64 × 35

4. 128 × 12

5. 216 × 30

Challenge NOT GRADED

C1. 216 × 34

C2. 347 × 26

The Standard Algorithm

Stack it, but keep an estimate in your head.

✦ Warm-Up 6 ITEMS

14×12

25×11

32×21

18×15

43×13

52×12

◆ Core GRADED • SHOW YOUR WORKING

1. 237×45

2. 418×27

3. 506×34

4. 725×18

5. 1234×12

★ Challenge NOT GRADED

C1. 2450×36

.....

C2. 3081×47

.....

Estimate to Check

Round both numbers, multiply, compare. A wrong answer should look wrong.

 Warm-Up 6 ITEMS

Estimate 19×21 by rounding both

Estimate 48×52

Estimate 31×29

Estimate 39×11

Estimate 62×18

Estimate 78×22

 Core GRADED • SHOW YOUR WORKING

1. Estimate 412×19

.....
2. Estimate 289×31

.....
3. True value of 412×19

.....
4. Estimate $5,120 \times 48$

.....
5. Estimate 197×203

.....

 Challenge NOT GRADED

C1. Someone says $34 \times 26 = 68$. Type the real answer (They multiplied by 2, not 26)

.....

C2. Someone says $250 \times 40 = 1000$. Type the real answer

.....

Poker & Floor Plan

Enrichment. Area two ways, and they must agree.

★ Warm-Up 2 ITEMS

Area of a 12 by 10 room, in squares

Area of a 20 by 15 room

◆ Core GRADED • SHOW YOUR WORKING

1. A 14 by 12 room scaled by 3: new length

2. That scaled room's area (42 by 36)

★ Challenge NOT GRADED

C1. Digits 3, 4, 7, 2 — biggest product from two 2-digit numbers: type it (73 × 42)

C2. Area of a 26 by 34 room

C3. Double both sides of a 15 by 20 room. Area multiplies by

Week 1 Answers

Mission 02 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 • Multiples of Ten

OUT OF 11

Warm-Up • 600 | 2000 | 4200 | 9000 | 1600 | 3600

Core • 24000 | 560000 | 1000 | 6000 | 7200

Challenge • 100000 | 100000

1.2 • Partial Products

OUT OF 11

Warm-Up • 230 | 92 | 322 | 322 | 620 | 62

Core • 682 | 1035 | 2240 | 1536 | 6480

Challenge • 7344 | 9022

1.3 • The Standard Algorithm

OUT OF 11

Warm-Up • 168 | 275 | 672 | 270 | 559 | 624

Core • 10665 | 11286 | 17204 | 13050 | 14808

Challenge • 88200 | 144807

1.4 • Estimate to Check

OUT OF 11

Warm-Up • 400 | 2500 | 900 | 400 | 1200 | 1600

Core • 8000 | 9000 | 7828 | 250000 | 40000

Challenge • 884 | 10000

Fri • Poker & Floor Plan

OUT OF 4

Warm-Up • 120 | 300

Core • 42 | 1512

Challenge • 3298 | 884 | 4

Multiples of Ten First

40 × 60 before 43 × 67. Every time.

Warm-Up 6 ITEMS

20×30

40×60

50×50

70×30

80×40

90×90

Core GRADED • SHOW YOUR WORKING

1. 400×70

2. 600×500

3. 250×40

4. 120×60

5. 35×200

Challenge NOT GRADED

C1. 1250×80

.....

C2. 2500×400

.....

Estimate, Then Compute

Round both, multiply, then do it properly and compare.

✦ Warm-Up 6 ITEMS

Estimate 19×21

Estimate 48×52

Estimate 31×29

Estimate 62×18

Estimate 78×22

Estimate 39×11

◆ Core GRADED • SHOW YOUR WORKING

1. Estimate 412×19

.....

2. True value of 412×19

.....

3. Estimate 289×31

.....

4. True value of 289×31

.....

5. Estimate 197×203

.....

★ Challenge NOT GRADED

C1. Estimate $5,120 \times 48$

.....

C2. Someone says $34 \times 26 = 68$. The real answer

.....

Four Digits by Two

Longer, not harder. Keep the estimate in your head.

 Warm-Up 6 ITEMS

1000×10

2000×20

1200×10

3000×30

1500×20

2500×10

 Core GRADED • SHOW YOUR WORKING

1. 1234×12

2. 2450×36

3. 1875×24

4. 3081×47

5. 4206×15

 Challenge NOT GRADED

C1. 5678×43

.....

C2. 9999×99

.....

Catch the Error

Given a wrong worked answer, find the line it went wrong.

 Warm-Up 6 ITEMS

23 × 14

23 × 4

23 × 10

92 + 230

31 × 22

45 × 23

 Core GRADED • SHOW YOUR WORKING

1. Someone got $23 \times 14 = 92$. What did they forget — type the missing partial
2. Someone got $45 \times 23 = 135$. The real answer
3. Someone got $38 \times 24 = 152$. The real answer
4. Someone got $56 \times 27 = 392$. The real answer
5. 63×48

 Challenge NOT GRADED

C1. 237×45

C2. Someone got $237 \times 45 = 1185$. The missing partial product

Mid-Unit Quiz

Eight items across Weeks 1–2. 85% to keep flying.

✦ Warm-Up 2 ITEMS

40×60

23×14

◆ Core GRADED • SHOW YOUR WORKING

1. 45×23

2. 237×45

3. Estimate 412×19

4. 1234×12

5. 506×34

★ Challenge NOT GRADED

C1. 2450×36

.....

Week 2 Answers

Mission 02 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 • Multiples of Ten First

OUT OF 11

Warm-Up • 600 | 2400 | 2500 | 2100 | 3200 | 8100

Core • 28000 | 300000 | 10000 | 7200 | 7000

Challenge • 100000 | 1000000

2.2 • Estimate, Then Compute

OUT OF 11

Warm-Up • 400 | 2500 | 900 | 1200 | 1600 | 400

Core • 8000 | 7828 | 9000 | 8959 | 40000

Challenge • 250000 | 884

2.3 • Four Digits by Two

OUT OF 11

Warm-Up • 10000 | 40000 | 12000 | 90000 | 30000 | 25000

Core • 14808 | 88200 | 45000 | 144807 | 63090

Challenge • 244154 | 989901

2.4 • Catch the Error

OUT OF 11

Warm-Up • 322 | 92 | 230 | 322 | 682 | 1035

Core • 230 | 1035 | 912 | 1512 | 3024

Challenge • 10665 | 9480

Fri • Mid-Unit Quiz

OUT OF 7

Warm-Up • 2400 | 322

Core • 1035 | 10665 | 8000 | 14808 | 17204

Challenge • 88200

Round to Estimate

Which way to round, and what it costs you.

✦ Warm-Up 6 ITEMS

Estimate 19×21

Estimate 48×52

Estimate 31×29

Estimate 62×18

Estimate 78×22

Estimate 39×11

◆ Core GRADED • SHOW YOUR WORKING

1. Estimate 412×19

.....

2. True value of 412×19

.....

3. Estimate 289×31

.....

4. True value of 289×31

.....

5. Estimate 197×203

.....

★ Challenge NOT GRADED

C1. Estimate $5,120 \times 48$

.....

C2. Estimate $2,450 \times 36$

.....

Over or Under

Say before you compute whether your estimate is high or low.

Warm-Up 6 ITEMS

Rounding both numbers up gives an estimate that is — over or under

Rounding both down gives — over or under

19×21

48×52

20×30

40×60

Core GRADED • SHOW YOUR WORKING

1. Estimate 412×19 as 400×20

.....

2. True value

.....

3. Was the estimate over or under

.....

4. Estimate 78×22 as 80×20

.....

5. True value of 78×22

.....

Challenge NOT GRADED

C1. 63×48

.....

C2. Estimating that as $60 \times 50 = 3000$ — over or under

.....

Order of Magnitude

Is it hundreds, thousands or tens of thousands?

Warm-Up 6 ITEMS

30×40

300×40

300×400

3×4

30×4

3000×4

Core GRADED • SHOW YOUR WORKING

1. Estimate 34×26 to the nearest hundred

.....

2. Estimate 340×26

.....

3. Estimate $3,400 \times 26$

.....

4. Digits in the answer to 34×26

.....

5. Digits in the answer to 340×26

.....

Challenge NOT GRADED

C1. Someone says $34 \times 26 = 68$. The real answer

.....

C2. Someone says $237 \times 45 = 1,185$. The real answer

.....

Mark Somebody's Work

Six worked answers, three wrong. Find them by estimating.

Warm-Up 6 ITEMS

23×14

45×23

31×22

38×24

56×27

12×12

Core GRADED • SHOW YOUR WORKING

1. 23×14 given as 92 — the missing partial product

.....

2. 45×23 given as 135 — the missing partial product

.....

3. 237×45 given as 1,185 — the missing partial product

.....

4. 56×27 with the second row as 112 — it should be

.....

5. That answer is out by

.....

Challenge NOT GRADED

C1. 78×46

C2. 94×68

Stadium Count Begins

Estimate from the photo before looking anything up.

Warm-Up 6 ITEMS

24 seats × 38 rows _____	20 × 25 _____	500 × 16 _____
30 × 30 _____	40 × 25 _____	1000 × 16 _____

Core GRADED • SHOW YOUR WORKING

1. 38 × 24	_____
2. 912 × 16	_____
3. Estimate 40 × 25 × 16	_____
4. The gap between that estimate and the true count	_____
5. 38 × 24 × 8	_____

Challenge NOT GRADED

C1. 2,450 × 36	_____
C2. 1,875 × 24	_____

Week 3 Answers

Mission 02 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 • Round to Estimate

OUT OF 11

Warm-Up • 400 | 2500 | 900 | 1200 | 1600 | 400

Core • 8000 | 7828 | 9000 | 8959 | 40000

Challenge • 250000 | 90000

3.2 • Over or Under

OUT OF 11

Warm-Up • over | under | 399 | 2496 | 600 | 2400

Core • 8000 | 7828 | over | 1600 | 1716

Challenge • 3024 | under

3.3 • Order of Magnitude

OUT OF 11

Warm-Up • 1200 | 12000 | 120000 | 12 | 120 | 12000

Core • 900 | 9000 | 90000 | 3 | 4

Challenge • 884 | 10665

3.4 • Mark Somebody's Work

OUT OF 11

Warm-Up • 322 | 1035 | 682 | 912 | 1512 | 144

Core • 230 | 900 | 9480 | 1120 | 1008

Challenge • 3588 | 6392

Fri • Stadium Count Begins

OUT OF 11

Warm-Up • 912 | 500 | 8000 | 900 | 1000 | 16000

Core • 912 | 14592 | 16000 | 1408 | 7296

Challenge • 88200 | 45000

Count a Section

Seats per row, rows per section, sections per stadium.

Warm-Up 6 ITEMS

24 × 2

24 × 10

24 × 20

24 × 30

24 × 38

12 × 24

Core GRADED • SHOW YOUR WORKING

1. 26 × 42

2. 32 × 48

3. 28 × 36

4. 30 × 45

5. 22 × 54

Challenge NOT GRADED

C1. 38 × 24 × 4

C2. 38 × 24 × 16

Scale It Up

One section times the number of sections.

✦ Warm-Up 6 ITEMS

900×10

900×16

1000×16

500×16

900×2

900×4

◆ Core GRADED • SHOW YOUR WORKING

1. 912×16 _____

2. 912×8 _____

3. $1,092 \times 12$ _____

4. $1,536 \times 10$ _____

5. $1,008 \times 20$ _____

★ Challenge NOT GRADED

C1. 14,592 to the nearest thousand
_____C2. 912×20

Explain the Gap

Compare to the published capacity and account for the difference.

Warm-Up 6 ITEMS

16,000 – 14,592

900 – 884

15,000 – 14,592

16,000 – 15,000

1000 – 912

2500 – 2496

Core GRADED • SHOW YOUR WORKING

1. Estimating 38×24 as 40×25 — over by

.....

2. That gap multiplied by 16

.....

3. 1,408 as a percent of 14,592, to the nearest whole

.....

4. Estimating 197×48 as 200×50 — the estimate

.....

5. True value of 197×48

.....

Challenge NOT GRADED

C1. 197×203

C2. An estimate of 40,000 is over by

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

✦ Warm-Up 6 ITEMS

7×8

40×60

23×14

12×6

400×70

9×9

◆ Core GRADED • SHOW YOUR WORKING

1. 45×23 _____

2. 237×45 _____

3. 506×34 _____

4. $1,234 \times 12$ _____

5. Estimate 412×19 _____

★ Challenge NOT GRADED

C1. $9,999 \times 99$

C2. $2,450 \times 36$

Mission 02 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 2 ITEMS

50 × 50

38 × 24

Core GRADED • SHOW YOUR WORKING

1. 56 × 27
2. 289 × 31
3. 125 × 24
4. 2,450 × 36
5. Estimate 5,120 × 48
6. Estimate 197 × 203
7. 1,875 × 24
8. 9,999 × 99

Challenge NOT GRADED

- C1. 237 × 45 given as 1,185 — the missing partial product
- C2. 38 rows of 24 seats in 16 sections

Week 4 Answers

Mission 02 • all 5 sets. Answers run in page order. Score out of the graded tiers only
— Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 • Count a Section

OUT OF 11

Warm-Up • 48 | 240 | 480 | 720 | 912 | 288

Core • 1092 | 1536 | 1008 | 1350 | 1188

Challenge • 3648 | 14592

4.2 • Scale It Up

OUT OF 11

Warm-Up • 9000 | 14400 | 16000 | 8000 | 1800 | 3600

Core • 14592 | 7296 | 13104 | 15360 | 20160

Challenge • 15000 | 18240

4.3 • Explain the Gap

OUT OF 11

Warm-Up • 1408 | 16 | 408 | 1000 | 88 | 4

Core • 88 | 1408 | 10 | 10000 | 9456

Challenge • 39991 | 9

Thu • Error Journal Sweep

OUT OF 11

Warm-Up • 56 | 2400 | 322 | 72 | 28000 | 81

Core • 1035 | 10665 | 17204 | 14808 | 8000

Challenge • 989901 | 88200

Fri • Mission 02 Test

OUT OF 10

Warm-Up • 2500 | 912

Core • 1512 | 8959 | 3000 | 88200 | 250000 | 40000 | 45000 | 989901

Challenge • 9480 | 14592